THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The average marks of 24 students is 42 and of another 18 students is 56. The average of all 42 students is: 1

(a) 50 **(b)** 47 **(c)** 49 **(d)** 48

2. A 600 mL mixture has 15% colour. The colour present is: 1

(a) 60 mL **(b)** 90 mL **(c)** 9 mL **(d)** 15 mL

3. In a stacked bar chart, the length of the whole bar shows: 1

(a) only the largest component **(b)** the percentage of the first component
(c) the average of the components **(d)** the total of all the components

4. Marks of 72, 76 and 88 are given weights 3, 2 and 3 respectively. The weighted average is: 1

(a) 80 **(b)** 81 **(c)** 78 **(d)** 79

5. The mean of 41, 13, 52, 42 is: 1

(a) 38 **(b)** 36 **(c)** 37 **(d)** 39

6. Over 3 weeks of equal length a cyclist's weekly average distances were 37, 24, 29 km per day. The average over the whole period (km per day) is: 1

(a) 30 **(b)** 29 **(c)** 31 **(d)** 32

7. In a 100% stacked bar of a household's monthly electricity use, appliances take 25% of 300 units. The units used by the appliances are: 1

(a) 74 **(b)** 77 **(c)** 75 **(d)** 76

8. A stacked bar for a family's monthly food use shows rice 12 kg, wheat 39 kg, pulses 29 kg. The total length of the bar is (in kg): 1

(a) 80 **(b)** 79 **(c)** 81 **(d)** 82

9. The average marks of 28 students is 59 and of another 14 students is 47. The average of all 42 students is: 1

(a) 56 **(b)** 55 **(c)** 57 **(d)** 54

10. A 800 g mixture has 35% nuts. The nuts present are: 1

(a) 280 g **(b)** 35 g **(c)** 28 g **(d)** 350 g

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A 300 mL bottle of a drink contains 19% fruit pulp. Find the volume of pulp and of the remaining liquid. 2

12. Find the mean of the numbers 70, 21, 69, 44. 2

13. A 100% stacked bar shows that 35% of a family's monthly spending of ₹160 is on food. Find the amount spent on food and on everything else. 2

14. A 350 mL bottle of a drink contains 24% fruit pulp. Find the volume of pulp and of the remaining liquid. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The average marks of 18 students in a test is 64 and that of 12 students is 49. Find the average marks of all 30 students together. 3

16. A girl cycles for 3 weeks of equal length. Her weekly averages are 53 min, 32 min, 50 min per day. Find the average over all the weeks and say why the simple mean of the averages can be used. 3

17. A final score combines Maths (score 40, weight 2), Science (score 85, weight 4). Find the weighted average score. 3

OR

The average marks of 25 students in a test is 30 and that of 20 students is 48. Find the average marks of all 45 students together. 3

18. A girl cycles for 4 weeks of equal length. Her weekly averages are 25 min, 53 min, 28 min, 26 min per day. Find the average over all the weeks and say why the simple mean of the averages can be used. 3

OR

A final score combines Maths (score 57, weight 4), Science (score 91, weight 4). Find the weighted average score. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of 5 observations is 29. A new observation 59 is added. Find the new mean. 5

OR

30 mL of a 22% sugar solution is mixed with 60 mL of a 19% sugar solution. Find the percentage concentration of the mixture. 5

20. The mean of 6 observations is 47. A new observation 61 is added. Find the new mean. 5

OR

20 mL of a 30% sugar solution is mixed with 40 mL of a 24% sugar solution. Find the percentage concentration of the mixture. 5

END OF QUESTION PAPER — Total: 40 Marks